

APPENDIX B—STRESS CONCENTRATION FACTORS FOR STRESSES DUE TO EXTERNAL LOADS

B.1 Introduction and Terminology

Presently available analytical methods for stresses in nozzles, shells, etc., do not take into account the localized stresses in fillets and transitions. The following data may prove useful in performing a fatigue analysis where such effects must be considered. In presenting these data, the following terminology will be used:

- H = thickness of thicker section of stepped bar, in.
- h = thickness of thinner section of stepped bar, in.
- r = fillet radius between two sections of bar, or between nozzle and shell, in.
- d_i = inside diameter of nozzle, in.
- d_n = outside diameter of nozzle, in. (see Fig. B-4).
- d_p = outside diameter of attached pipe, in. (see Fig. B-4).
- t = thickness of pipe, in.
- t^1 = thickness of reinforced section of nozzle, in. (see Fig. B-4).
- T = thickness of vessel wall, in.
- θ = angle of taper between two sections of bar or nozzle, degrees (see Figs. B-3 and B-4).
- K_T = stress concentration factor at fillet of a stepped bar in tension (as related to the stress in the *thinner* member)
- K_B = ditto, for a stepped bar in bending (as related to the surface stress of the *thinner* member)
- K_n = stress concentration factor applied to the membrane portion of the stress due to external nozzle loadings ("tension" curve on Fig. B-2; see para. B.3.1)
- K_b = stress concentration factor applied to the bending portion of the stress due to external nozzle loadings ("bending" curve on Fig. B-2; see para. B.3.1)
- K_0 = stress concentration factor at fillet of stepped bar for case of $\theta = 0^\circ$
- K^1 = stress concentration factor for inclined shoulder

B.2 Stress Concentration Factors

Peterson²⁹ and Heywood³⁰ provide a considerable amount of data covering the stress concentration factors for various design problems including the cases of two-dimensional, stepped bars in both tension and bending. In both cases, curves are provided giving stress concentration factors in terms of the H/h^* ratio of the bar and the fillet

radius between the two sections of the bar, as illustrated in Fig. B-1. Heywood's curves conform to the following formulas:

Tension case

$$K_T = 1 + \left[\frac{\frac{H}{h} - 1}{4 \left(1.4 \frac{H}{h} - 1 \right)} \times \frac{h}{r} \right]^{0.65} \quad (1)$$

Bending case

$$K_B = 1 + \left[\frac{\frac{H}{h} - 1}{2 \left(5.37 \frac{H}{h} - 4.8 \right)} \times \frac{h}{r} \right]^{0.85} \quad (2)$$

For the case of a plate of "infinite" width (H very large in relation to h), these formulas would reduce to:

$$\text{Tension} \quad K_T = 1 + \left[\frac{1}{5.6 \frac{r}{h}} \right]^{0.65} \quad (3)$$

$$\text{Bending} \quad K_B = 1 + \left[\frac{1}{10.74 \frac{r}{h}} \right]^{0.85} \quad (4)$$

Upon comparing Heywood's computed curves with the corresponding curves from Peterson,²⁹ it will be found that the tension curves are quite consistent with Peterson's data, but that the bending curves are seemingly somewhat "low" for small values of r/h . The following alternate formula provides a curve which is more consistent with Peterson's data (and somewhat conservative in relation to Heywood's data):

$$K_B = 1 + \left[\frac{1}{9.4 \frac{r}{h}} \right]^{0.80} \quad (5)$$

Curves based on formulas (3) and (5) are shown on Fig. B-2, taking H as infinite and taking h as equal to $2T$ when applied to the vessel shell and equal to

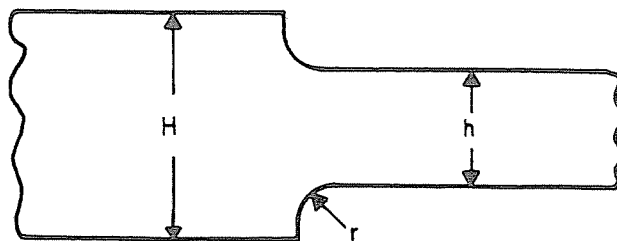


Fig. B-1—Stepped bar

* Terminology changed to avoid conflict with terminology in this text.

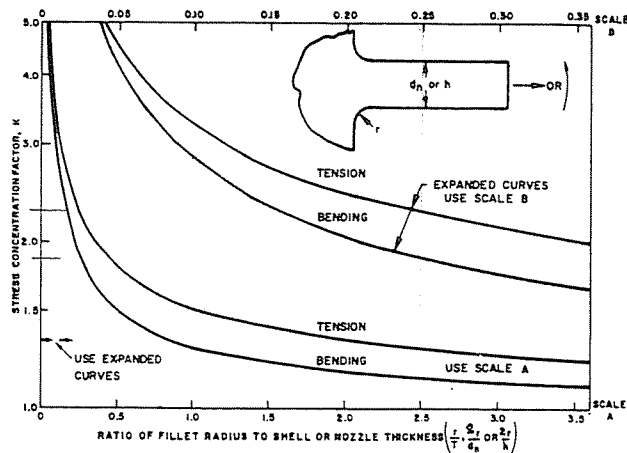


Fig. B-2—Stress concentration factors for $D \gg d$
or $H \gg h$

d_n when applied to the nozzle* (see Fig. B-4).

The case of an inclined shoulder is also of interest, as discussed on page 179 of Reference 30. If the stress concentration factor obtained from Fig. 57 or 60 of Reference 29 is designated K_0 , and the stress concentration factor for the inclined shoulder as K' , the following relationships apply:

- (1) If $r < (t' - t)$

$$\frac{K' - 1}{K_0 - 1} = 1 - \left[\frac{\theta}{90} \right]^n \quad (6)$$

where θ is the angle between the tapered shoulder and the square shoulder (see Fig. B-3), and $n = 1 + 2.4 \sqrt{r/(t' - t)}$

- (2) If $r > h$

$$\frac{K' - 1}{K_0 - 1} = 1 - \left[\frac{\theta - \alpha}{90} \right]^n \quad (7)$$

where

$$\alpha = \sin^{-1} \left[1 - \frac{(t' - t)}{r} \right]$$

- (3) If $H \gg h$ for use with Fig. B-2

$$\frac{K' - 1}{K_0 - 1} = 1 - \frac{\theta}{90} \quad (8)$$

The quantities on the left-hand side of the above eqs. 6, 7, and 8 are plotted in Fig. B-3 vs. the quantity of the right-hand side.

B.3 Application to External Loads on a Nozzle

B.3.1 Stresses at Fillet-Shell Junction

The calculated stresses in the shell due to an axial load or moment loading on a nozzle are assumed to apply to the juncture of the fillet and shell, point A in Fig. B-4. These stresses have been derived in terms of a membrane component (N/T) and a bending component ($6M/T^2$), in the

* Since less data are available for stress concentration factors in shafts than in bars or plates, Heywood (Reference 30, page 195) recommended that two-dimensional plate data be used. In general, it is believed that such data are slightly conservative in relation to three-dimensional data.

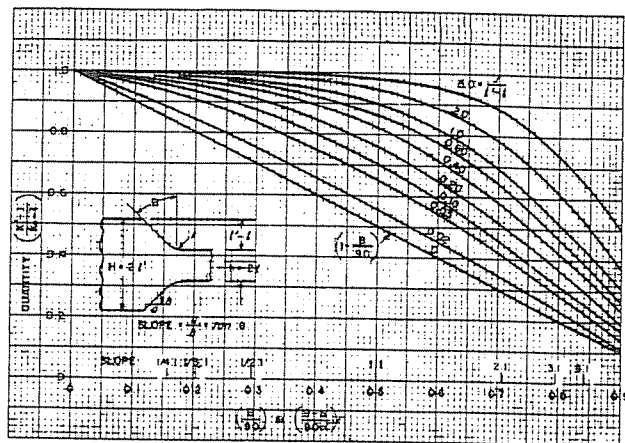


Fig. B-3—Effect of tapered shoulder

form:

$$\sigma_i = K_n \frac{Ni}{T} \pm K_b \frac{6Mi}{T^2}$$

The stress concentration factor, K_n , for the membrane component is that obtained from Fig. B-2 for the tensile case, using the appropriate ratio of fillet radius-to-plate thickness, r_A/T (see Fig. B-4). The factor, K_b , for the bending component is determined from the same figure for the bending case.

In both cases, the stress concentration factor is applied to the stresses which are *perpendicular to the change in section*. In the normal case, these are the stresses which are oriented *axially with respect to the nozzle*, i.e., σ_z in the case of a spherical vessel or the longitudinal axis of a cylindrical vessel, and σ_ϕ in the case of the transverse axis of a cylindrical vessel.

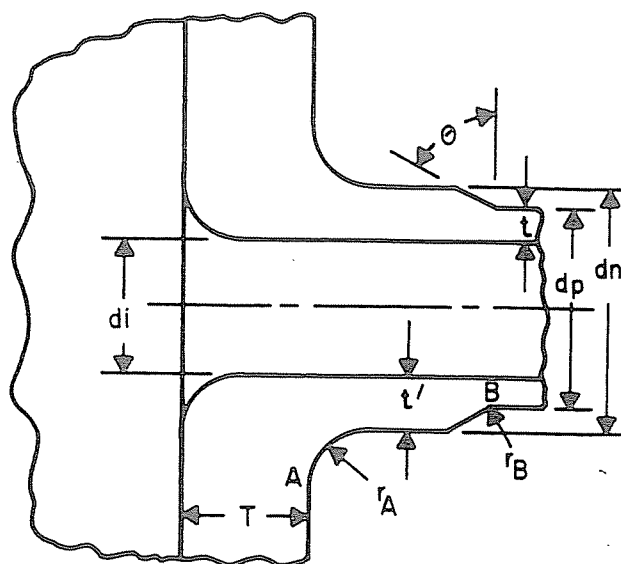


Fig. B-4—Nozzle configuration

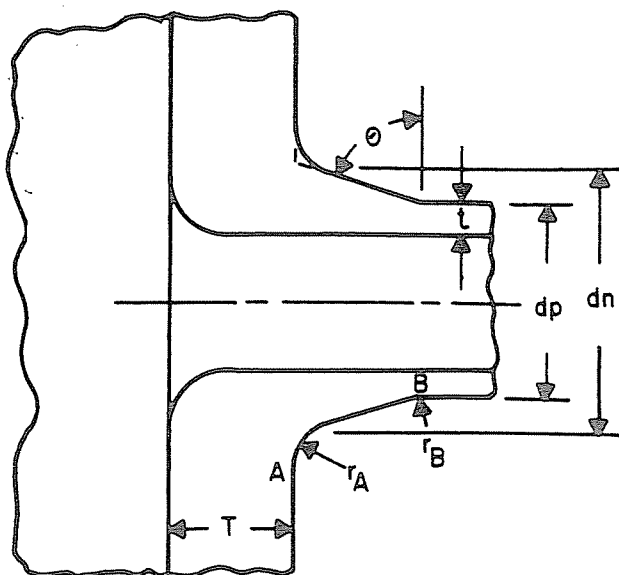


Fig. B-5—Nozzle configuration

B.3.2 Stresses at Nozzle-Pipe Junction

The stress concentration factor for an axial load, or for a bending moment, at the juncture of a nozzle and its attached pipe (Point B in Fig. B-4) may be obtained from Figs. 57 and 60 of Reference 29* and Fig. B-3 herein in terms of d_n , d_p , r_B and θ . These factors should be applied to the stress in the pipe (thinner member) calculated from the conventional P/A and Mc/I relationships.

B.3.3 Special Case of a Tapered Nozzle

The following procedure should be used for a special case of a tapered nozzle as shown in Fig. B-5:

- Calculate the stress concentration factor that would exist at Point A if the nozzle were of uniform diameter, d_n .
- Account for the taper by using the quantity $(90 - \theta)$ in place of θ in the abscissa of Fig. B-3.
- The stress at the nozzle-pipe junction, point B, should be calculated in the same manner as in para. B.3.2.

B.4 Application to Bars and Structural Attachments

The stress concentration at the juncture of a

* When using data from Reference 29, changes in symbols as contained in this text should be noted.

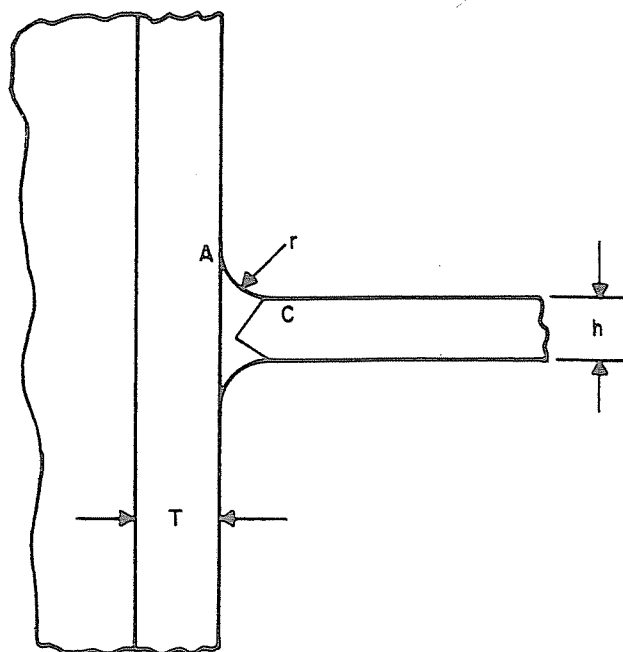


Fig. B-6—Structural attachment configuration

structural attachment to a shell, as shown in Fig. B-6, may be treated in the same way as for a nozzle, as covered in para. B.3.1. Another location on the attachment which may be critically stressed is point 'C', at the juncture of the fillet and the attachment. The stresses at this point may be calculated by applying the appropriate factor from Fig. B-2 to the conventionally calculated P/A and Mc/I stresses in the bar or attachment. [NOTE: This procedure can also be applied to a nozzle provided that the nozzle is relatively "rigid" (thick in relation to its diameter). However, experimental data indicate that such treatment would not be proper for relatively thin, flexible nozzles, which have high bending stresses *through the thickness* of the nozzle as differentiated from the beam action of a thick-walled nozzle.]

B.5 Acknowledgment

The foregoing material constitutes an adaptation of material originally presented in Reference 16, with added data documenting the genesis of the curves shown in Fig. B-2.

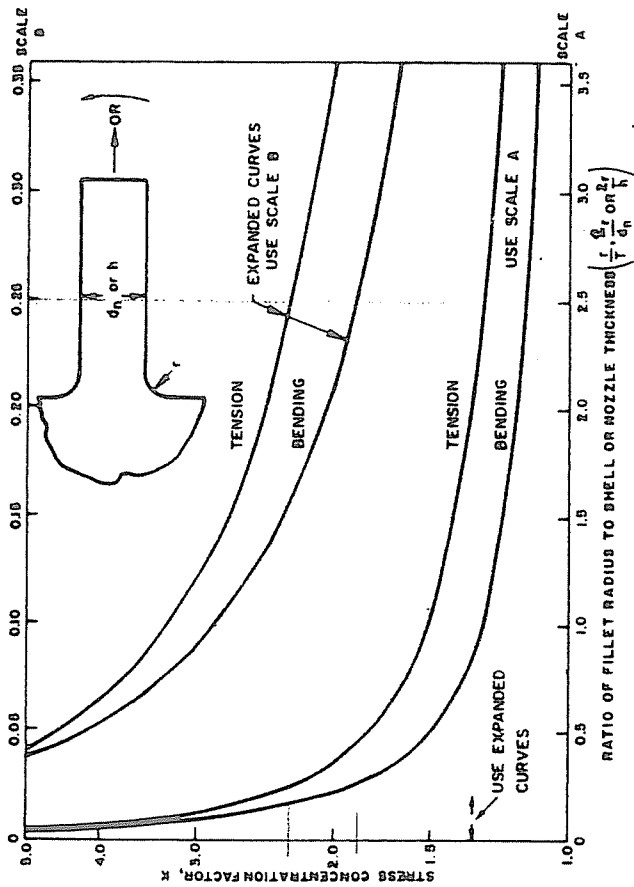


Fig. B-2—Stress concentration factors for $D \gg d$
OR $H \gg h$,

d_n when applied to the nozzle* (see Fig. B-4).

The case of an inclined shoulder is also of interest, as discussed on page 179 of Reference 30. If the stress concentration factor obtained from Fig. 57 or 60 of Reference 29 is designated K_0 , and the stress concentration factor for the inclined shoulder as K' , the following relationships apply:

$$(1) \quad \text{If } r < (t' - t)$$

$$\frac{K' - 1}{K_0 - 1} = 1 - \left[\frac{\theta}{90} \right]^n \quad (6)$$

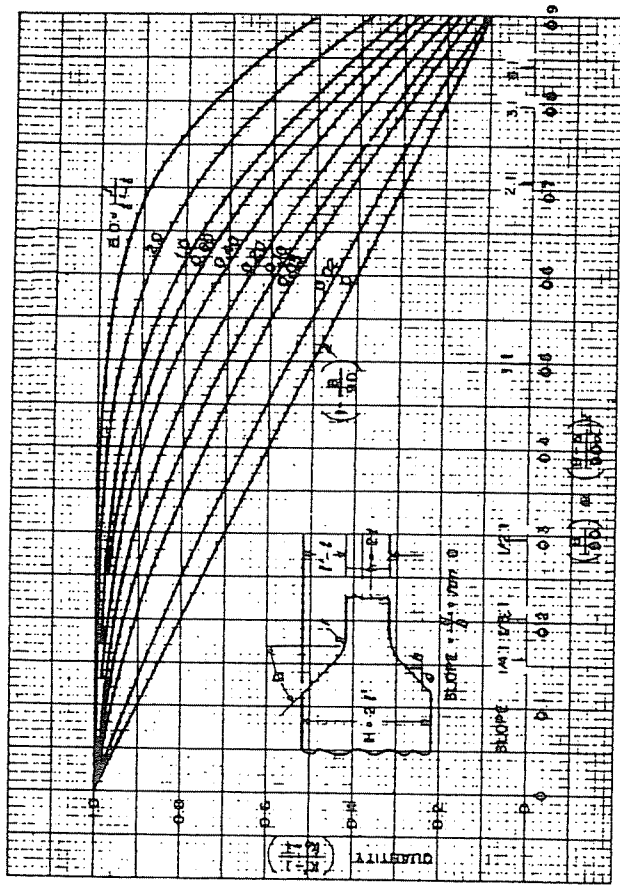


Fig. B-3—Effect of tapered shoulder

form:

$$\sigma_t = K_n \frac{Ni}{T} \pm K_b \frac{6Mi}{T^2}$$

The stress concentration factor, K_n , for the membrane component is that obtained from Fig. B-2 for the tensile case, using the appropriate ratio of fillet radius-to-plate thickness, r_A/T (see Fig. B-4). The factor, K_b , for the bending component is determined from the same figure for the bending case.